

How to improve a neural network:-

If u have an Art. neural network
so how can u increase performance.

1) Fine tuning of hyperparameters

like hidden layers no of neurons,

Learning rate, optimizer, Batch size, Act func

Epochs.

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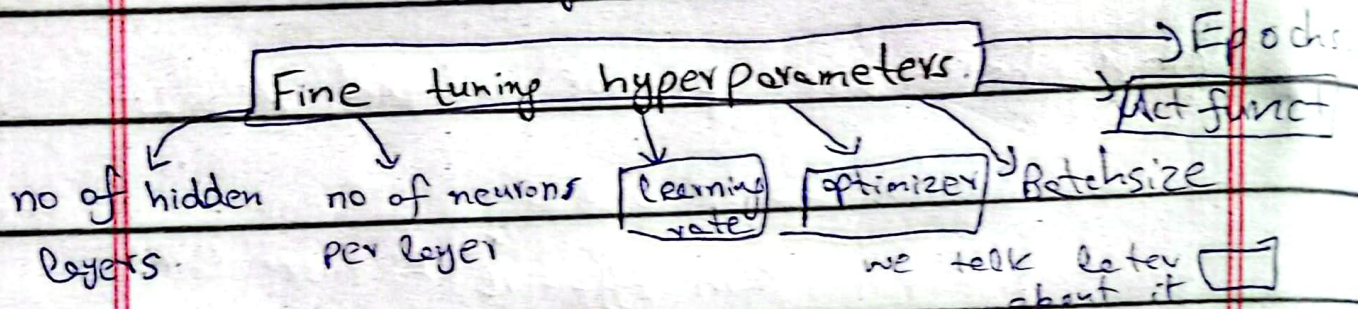
2:- By solving problems

→ Vanishing/exploding gradient

→ Not enough data.

→ Slow training.

→ Overfitting.



a) No of hidden layers:-

Take more hidden layers with fewer neurons rather than taking 1 layer with a lot of neuron.

→ increase hidden layers till overfitting.

→ Representation learning is that let's

say we have a face  so 1st layer

capture - o pattern, 2nd complex, last full face, this is R. learning.

primitive layer:- capture patterns.

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Neuron/layer:- 1st approach was
of pyramid 

→ but now , no of neurons
per layer must be sufficient.

c) Batch size:-

How would u decide batch.

smaller (8 to 32) larger. (training fast)
(8 to 32) (8 to 2)

↓
better result
on new data

↓
Now if u take good
& improved result then
this is better so some
ppl worked on this.

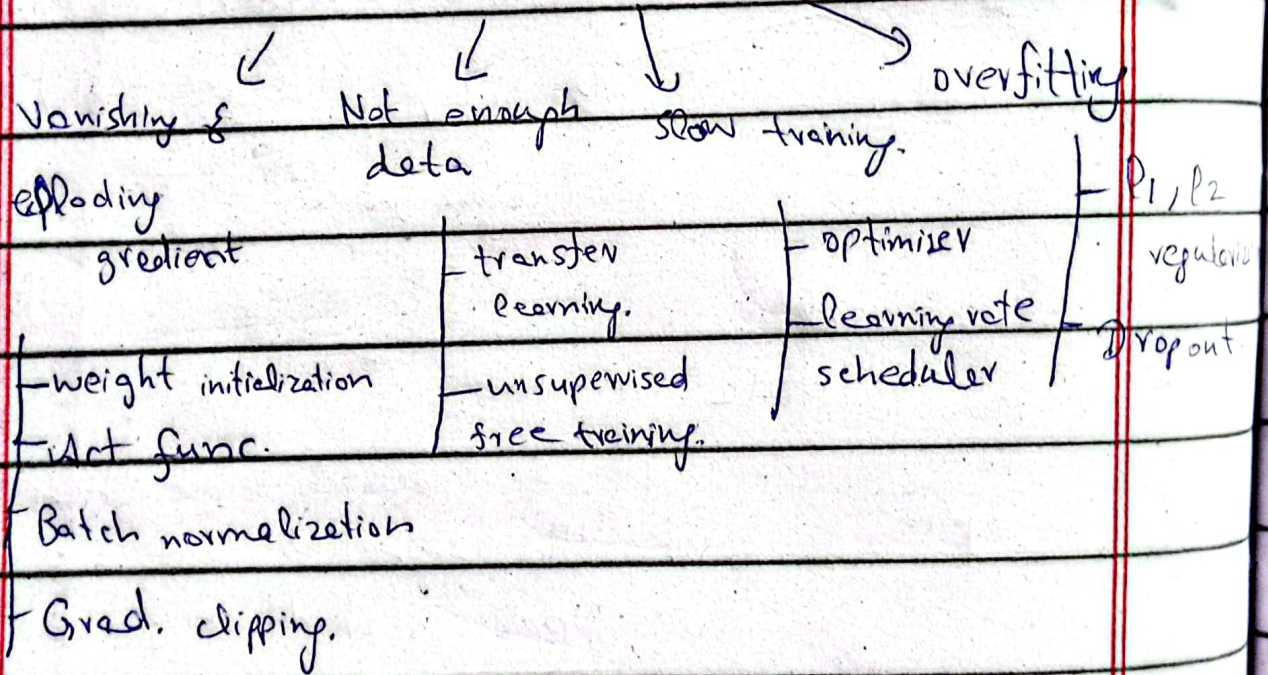
Learning rate scheduler.
means warming up, l. rate

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d) epochs:- early stopping
↓
keras → stable.
mechanism which is intelligent enough
keras takes care of it.
keras → call back func.

Problems with N.N.



Early stopping:-

E.S is a mechanism in keras that automatically detects that after a point if u train

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your model isn't beneficial instead
it's a loss.

→ In Keras early stopping implemented
through `callbacks` feature.

Early Stopping (`EarlyStopping`) loss on new data.

`monitor = "val_loss"`, → I can do with accuracy

`min_delta = 0.01`, → min change in monitored quantity
to verify as improvement.

`patience = 20`, → No of epochs with no improvement
after which training will stop.

`verbose = 0`,

`mode = auto`,

`baseline = None`,

`restore_best_weights = False`

)